*Acta Sci. Math.* (*Szeged*)
**64**/3–4 (1998), 495–513

# Partial-fraction expansions and well-poised bilateral series

Chu Wenchang*

*Communicated by P. Hajnal*

**Abstract.** As a common generalization of the formulas due to Minton (1970), Karlsson (1971) and Chu (1994), a well-poised bilateral hypergeometric identity is established by means of partial-fraction expansions. The basic counterpart is also obtained by the Cauchy residue method. Several terminating cases are considered and dualized, by using the inverse series relations of Chu (1993), which results in some strange hypergeometric evaluations.

## 1. Introduction and notation

For complex numbers $x$ and $q$, the shifted-factorial of order $n$ and its $q$-analogue are defined, respectively, by

$$(x)_n = x\,(x+1)\cdots(x+n-1) \tag{1.1a}$$

$$= \Gamma(x+n)/\Gamma(x) \tag{1.1b}$$

$$(x;\,q)_n = (1-x)\,(1-xq)\cdots(1-xq^{n-1}) \tag{1.1c}$$

$$= (x;\,q)_\infty/(xq^n;\,q)_\infty \tag{1.1d}$$

where $n$ is assumed, respectively, to be a nonnegative integer for (1.1a) and (1.1c), and integral for (1.1b) and (1.1d).

Their fractions are abbreviated, in order to save writing space, to

$$\Gamma\begin{bmatrix} a, & b, & \cdots, & c \\ A, & B, & \cdots, & C \end{bmatrix} = \frac{\Gamma(a)\,\Gamma(b)\cdots\Gamma(c)}{\Gamma(A)\,\Gamma(B)\cdots\Gamma(C)} \tag{1.2a}$$

Received December 15, 1997.
*AMS Subject Classification* (1991): 33A30, 05A19.
* Partially supported by Colomba & Yuang Foundation.

All rights reserved © Bolyai Institute, University of Szeged

$$\begin{bmatrix} a, & b, & \cdots, & c \\ A, & B, & \cdots, & C \end{bmatrix}_n = \frac{(a)_n (b)_n \cdots (c)_n}{(A)_n (B)_n \cdots (C)_n} \tag{1.2b}$$

$$\begin{bmatrix} a, & b, & \cdots, & c \\ A, & B, & \cdots, & C \end{bmatrix}_n = \frac{(a;\,q)_n\,(b;\,q)_n \cdots (c;\,q)_n}{(A;\,q)_n\,(B;\,q)_n \cdots (C;\,q)_n}. \tag{1.2c}$$

Following Bailey [1] and Slater [8], the unilateral and bilateral (ordinary and basic) hypergeometric series are defined, respectively, by

$${}_{1+r}F_s\left[\begin{matrix} a_0, & a_1, & \cdots, & a_r \\ & b_1, & \cdots, & b_s \end{matrix}; z\right] = \sum_{n=0}^{\infty} \frac{z^n}{n!} \begin{bmatrix} a_0, a_1, \cdots, a_r \\ b_1, \cdots, b_s \end{bmatrix}_n \tag{1.3a}$$

$${}_{1+r}\phi_s\left[\begin{matrix} a_0, & a_1, & \cdots, & a_r \\ & b_1, & \cdots, & b_s \end{matrix}; z\right] = \sum_{n=0}^{\infty} \frac{z^n}{(q;\,q)_n} \begin{bmatrix} a_0, a_1, \cdots, a_r \\ b_1, \cdots, b_s \end{bmatrix}_n \tag{1.3b}$$

$${}_{r}H_s\left[\begin{matrix} a_1, & a_2, & \cdots, & a_r \\ b_1, & b_2, & \cdots, & b_s \end{matrix}; z\right] = \sum_{n=-\infty}^{+\infty} z^n \begin{bmatrix} a_1, a_2, \cdots, a_r \\ b_1, b_2, \cdots, b_s \end{bmatrix}_n \tag{1.3c}$$

$${}_{r}\psi_s\left[\begin{matrix} a_1, & a_2, & \cdots, & a_r \\ b_1, & b_2, & \cdots, & b_s \end{matrix}; z\right] = \sum_{n=-\infty}^{+\infty} z^n \begin{bmatrix} a_1, a_2, \cdots, a_r \\ b_1, b_2, \cdots, b_s \end{bmatrix}_n \tag{1.3d}$$

where the unilateral series may be considered as special cases of the corresponding bilateral series. When the complex variable $z = 1$, it will be omitted from the ordinary hypergeometric notation.

Throughout the paper, $n_1, n_2, \ldots$, and $n_p$ are assumed invariably to be non-negative integers with sum $N = \sum_{k=1}^{p} n_k$. The base $q$ will be confined to $|q| < 1$ for non-terminating q-series.

For the hypergeometric series with integral parameter differences, an interesting formula found by Karlsson [6] may be restated as

$${}_{p+2}F_{1+p}\left[\begin{matrix} a, & d, & x_1+n_1, & x_2+n_2, & \cdots, & x_p+n_p \\ & 1+d, & x_1, & x_2, & \cdots, & x_p \end{matrix}\right] \tag{1.4a}$$

$$= \Gamma\begin{bmatrix} 1-a,\ 1+d \\ 1-a+d \end{bmatrix} \prod_{k=1}^{p} \frac{(x_k-d)_{n_k}}{(x_k)_{n_k}}, \qquad (N + \Re(a) < 1) \tag{1.4b}$$

which is a non-terminating extension of an earlier result due to Minton [7].

Recently, the author [4] has obtained its bilateral version

$${}_{p+2}H_{2+p}\left[\begin{matrix} a, & d, & x_1+n_1, & x_2+n_2, & \cdots, & x_p+n_p \\ c, & 1+d, & x_1, & x_2, & \cdots, & x_p \end{matrix}\right] \tag{1.5a}$$

$$= \Gamma\begin{bmatrix} 1-a,\ c \\ 1-a+d,\ c-d \end{bmatrix} \frac{\pi d}{\sin \pi d} \prod_{k=1}^{p} \frac{(x_k-d)_{n_k}}{(x_k)_{n_k}} \tag{1.5b}$$

$$(N + \Re(a-c) < 0)$$


All rights reserved © Bolyai Institute, University of Szeged

as well as the $q$-analogue

$$(1.6a)\qquad {}_{p+2}\psi_{2+p}\left[\begin{matrix} a, & d, & q^{n_1}x_1, & q^{n_2}x_2, & \cdots, & q^{n_p}x_p \\ c, & qd, & x_1, & x_2, & \cdots, & x_p \end{matrix}; q^{1-m}/a\right]$$

$$(1.6b)\qquad = \left[\begin{matrix} q, & q, & c/d, & qd/a \\ q/a, & c, & qd, & q/d \end{matrix}; q\right]_\infty \mathbf{d}^{\mathbf{m}} \prod_{k=1}^{p} \frac{(x_k/d;\, q)_{n_k}}{(x_k;\, q)_{n_k}}$$

$$(|q/a| < |q^m| < |q^N/c|)$$

which contains, in particular, the results in [5] on unilateral $q$-series as special cases.

The purpose of this paper is to generalize the formulae of Chu–Karlsson–Minton further to the very well-poised bilateral hypergeometric series, which may provide also an explanation for the existence of this class of hypergeometric evaluations. Then we go on naturally pursuing its $q$-analogue, a basic very well-poised bilateral identity. When the series considered terminate, we may examine several exceptional cases. Their dual formulas are demonstrated by means of the inversion technique developed by Chu [3] to be a strange hypergeometric evaluation and its q-series counterpart.

## 2. Bilateral hypergeometric evaluation

For complex numbers $a$, $b$, $c$ and $\{x_k, y_k\}$, suppose that $x_k + y_k = 1 + a + n_k$, where $\{n_k\}$ are nonnegative integers with $N = \sum n_k$. The meromorphic function in complex variable $z$, with additive reflection property $f(z) = f(a-z)$, defined by

$$(2.1a)\quad f(z) = \frac{\sin \pi a}{\pi a}\,\frac{\pi z}{\sin \pi z}\,\frac{\pi(a-z)}{\sin \pi(a-z)} \prod_{k=1}^{p} \left[\begin{matrix} 1+z-x_k, & 1+z-y_k \\ 1+a-x_k, & 1+a-y_k \end{matrix}\right]_{n_k} \times$$

$$(2.1b)\qquad \times \Gamma\left[\begin{matrix} 1+a-b, & 1+a-c, & 1-b, & 1-c \\ 1+a-b-z, & 1+a-c-z, & 1-b+z, & 1-c+z \end{matrix}\right]$$

has simple poles $z = j$ and $z = a + j$ $(j = \pm 1, \pm 2, \cdots)$ with residues

$$(2.2a)\quad z = -j, -j\frac{a+j}{a}\left[\begin{matrix} b, & c \\ 1+a-b, & 1+a-c \end{matrix}\right]_j \prod_{k=1}^{p} \left[\begin{matrix} x_k, & y_k \\ 1+a-x_k, & 1+a-y_k \end{matrix}\right]_j$$

$$(2.2b)\quad z = a + j, j\frac{a+j}{a}\left[\begin{matrix} b, & c \\ 1+a-b, & 1+a-c \end{matrix}\right]_j \prod_{k=1}^{p} \left[\begin{matrix} x_k, & y_k \\ 1+a-x_k, & 1+a-y_k \end{matrix}\right]_j$$

in accordance with

$$(2.3)\qquad \operatorname{Res}\left[\frac{1}{\sin \pi(z+\tau)}\right]_{z=n-\tau} = (-1)^n/\pi.$$


All rights reserved © Bolyai Institute, University of Szeged

Denote by $\{x_k, y_k\}$ the $n$-pairs of hypergeometric parameters for $1 \le k \le n$ and for the sake of brevity let $x^* = 1 + a - x$, then we have a partial-fraction expansion (cf. Titchmarsh [9, §3.2])

$$\text{(2.4a)}\quad {}_{2p+5}H_{5+2p}\left[\begin{matrix} 1+a/2, & b, & c, & z, & a-z, & \{x_k, y_k\} \\ a/2, & b^*, & c^*, & z^*, & 1+z, & \{x_k^*, y_k^*\} \end{matrix}\right]$$

$$\text{(2.4b)}\qquad \rightleftharpoons f(z) = f(0) + z\sum_{j\neq 0}\left\{\frac{\operatorname{Res}[f(z)]_{z=a+j}}{(a+j)\,(z-a-j)} - \frac{\operatorname{Res}[f(z)]_{z=-j}}{j\,(z+j)}\right\}$$

provided that there is a sequence of circles $\mathcal{C}_n$ with center at the origin of the complex plane, not passing any poles, such that there holds $f(z) = o(R_n)$ uniformly on the system of contours $\{\mathcal{C}_n\}$, where $R_n$ is the radius of $\mathcal{C}_n$.

For $|a| = \varepsilon(\mathrm{mod}\ 1)$, denote by $\mathcal{C}_n$ the circle with radius equal to $n + (1+\varepsilon)/2$ and center at the origin. Writing $a = \alpha + i\beta$ and $z = x + yi$, then there is a constant $M > 0$ such that the following inequalities are valid on $\mathcal{C}_n$.

$$\text{(2.5a)}\qquad |\sin \pi z| \ge 1/M \ne 0$$

$$\text{(2.5b)}\qquad |\sin \pi(a-z)| \ge 1/M \ne 0$$

$$\text{(2.5c)}\qquad |\operatorname{ctg} \pi z|^2 = \frac{\cos \pi z\ \cos \pi \bar{z}}{\sin \pi z\ \sin \pi \bar{z}}$$

$$\text{(2.5d)}\qquad = \left|\frac{\cos 2\pi x + \cos 2\pi y i}{\cos 2\pi x - \cos 2\pi y i}\right|$$

$$\text{(2.5e)}\qquad = \left|\frac{\cos 2\pi x + \cosh 2\pi y}{\cos 2\pi x - \cosh 2\pi y}\right|$$

$$\text{(2.5f)}\qquad < M^2(\text{bounded})$$

$$\text{(2.5g)}\qquad |\operatorname{ctg} \pi(a-z)|^2 = \left|\frac{\cos 2\pi(\alpha - x) + \cosh 2\pi(\beta - y)}{\cos 2\pi(\alpha - x) - \cosh 2\pi(\beta - y)}\right|$$

$$\text{(2.5h)}\qquad < M^2(\text{bounded})$$

$$\text{(2.5i)}\qquad \left|\frac{\sin \pi(\theta - z)}{\sin \pi z}\right| \le |\sin \pi\theta| \cdot |\operatorname{ctg} \pi z| + |\cos \pi\theta|$$

$$\text{(2.5j)}\qquad \le |\cos \pi\theta| + M\,|\sin \pi\theta|$$

$$\text{(2.5k)}\qquad \left|\frac{\sin \pi(\theta - z)}{\sin \pi(a - z)}\right| \le |\sin \pi(\theta - a)| \cdot |\operatorname{ctg} \pi(a-z)| + |\cos \pi(\theta - a)|$$

$$\text{(2.5l)}\qquad \le |\cos \pi(\theta - a)| + M\,|\sin \pi(\theta - a)|.$$

By means of the reciprocity of $\Gamma$-functions

$$\text{(2.6)}\qquad \Gamma(t)\,\Gamma(1-t) = \frac{\pi}{\sin \pi t}$$


All rights reserved © Bolyai Institute, University of Szeged

we may restate $f(z)$, according to its argument, as

$$f(z)=z\,(a-z)\frac{\sin\pi(b+z-a)\ \sin\pi(c+z-a)}{\sin\pi z\ \sin\pi(a-z)}\Gamma\left[\begin{matrix}b+z-a, & c+z-a\\ 1-b+z, & 1-c+z\end{matrix}\right]\times \tag{2.7}$$

$$\times\,\Gamma[4/0]\,\frac{\sin\pi a}{\pi a}\prod_{k=1}^{p}\left[\begin{matrix}1+z-x_k, & 1+z-y_k\\ 1+a-x_k, & 1+a-y_k\end{matrix}\right]_{n_k},\quad |\arg(z)|<\pi \tag{2.8}$$

$$=z\,(a-z)\frac{\sin\pi(b-z)\ \sin\pi(c-z)}{\sin\pi z\ \sin\pi(a-z)}\Gamma\left[\begin{matrix}b-z, & c-z\\ 1+a-b-z, & 1+a-c-z\end{matrix}\right]\times \tag{2.9}$$

$$\times\,\Gamma[4/0]\,\frac{\sin\pi a}{\pi a}\prod_{k=1}^{p}\left[\begin{matrix}1+z-x_k, & 1+z-y_k\\ 1+a-x_k, & 1+a-y_k\end{matrix}\right]_{n_k},\quad |\arg(-z)|<\pi \tag{2.10}$$

where

$$\Gamma[4/0]=\Gamma[1-b,\ 1-c,\ 1+a-b,\ 1+a-c]. \tag{2.11}$$

Applying the Stirling formula (cf. [9, §4.42]) to this reformulation with reference to the trigonometric estimations displayed in (2.5), we confirm that

$$f(z)=O(|z|^{2N-2a+2b+2c}),\, z\in\mathcal{C}_n.$$

When $N+\Re(b+c-a)<1/2$, we get identity (2.4). By means of analytic continuation, this condition may be replaced by $N+\Re(b+c-a)<1$ under which (2.4a) is also convergent.

Summing up, we may state our main result from (2.4) with $z$ being replaced by $d$, as a very well-poised bilateral hypergoemetric identity.

**Theorem 1.** *For complex numbers $a,b,c,d$ and $\{x_k,y_k\}_{k=1}^{p}$ satisfying $x_k+y_k=1+a+n_k$ $(k=1,2,\cdots,p)$ with $\{n_k\}_{k=1}^{p}$ being nonnegative integers and $N=\sum_{k=1}^{p}n_k$, there holds*

$${}_{2p+5}H_{5+2p}\left[\begin{matrix}1+a/2, & b, & c, & d, & a-d, & \{x_k,y_k\}\\ a/2, & b^*, & c^*, & d^*, & 1+d, & \{x_k^*,y_k^*\}\end{matrix}\right] \tag{2.12a}$$

$$=\frac{\sin\pi a}{\pi a}\,\frac{\pi d}{\sin\pi d}\,\frac{\pi(a-d)}{\sin\pi(a-d)}\prod_{k=1}^{p}\left[\begin{matrix}1+d-x_k, & 1+d-y_k\\ 1+a-x_k, & 1+a-y_k\end{matrix}\right]_{n_k}\times \tag{2.12b}$$

$$\times\,\Gamma\left[\begin{matrix}1+a-b, & 1+a-c, & 1-b, & 1-c\\ 1+a-b-d, & 1+a-c-d, & 1-b+d, & 1-c+d\end{matrix}\right] \tag{2.12c}$$

*provided that $N+\Re(b+c-a)<1$ and the bilateral series is well-defined.*


All rights reserved © Bolyai Institute, University of Szeged

The Chu–Karlsson–Minton formulae (1.4–1.5) may be regarded as its limiting case of $a \to \infty$ after replacing $c$ by $1+a-c$. When $N=0$, it reduces to a special case of the Dougall formula (cf. [8, §6.1])

$$
\begin{aligned}
&(2.13a)\quad {}_5H_5\left[\begin{matrix} 1+a/2, & b, & c, & d, & e \\ a/2, & 1+a-b, & 1+a-c, & 1+a-d, & 1+a-e \end{matrix}\right] \\
&(2.13b)\qquad = \Gamma\left[\begin{matrix} 1+a-b,\ 1+a-c,\ 1+a-d,\ 1+a-e \\ 1+a-b-c,\ 1+a-b-d,\ 1+a-c-d,\ 1-a \end{matrix}\right]\times \\
&(2.13c)\qquad\qquad \times \Gamma\left[\begin{matrix} 1-b,\ 1-c,\ 1-d,\ 1-e,\ 1+2a-b-c-d-e \\ 1+a-b-e,\ 1+a-c-e,\ 1+a-d-e,\ 1+a \end{matrix}\right]
\end{aligned}
$$

where $\Re(1+2a-b-c-d-e)>0$.

## 3. Basic very well-poised analogue

Recalling that many hypergeometric identities have their basic counterpart, we may state the $q$-analogue of Theorem 1 as follows.

**Theorem 2.** *For complex numbers $a,b,c,d$ and $\{x_k,y_k\}_{k=1}^p$ satisfying $x_k\cdot y_k = aq^{1+n_k}$ $(k=1,2,\dots,p)$ with $\{n_k\}_{k=1}^p$ being nonnegative integers and $N=\sum_{k=1}^p n_k$, there holds*

$$
\begin{aligned}
&(3.1a)\quad {}_{2p+6}\psi_{6+2p}\left[\begin{matrix} q\sqrt{a}, & -q\sqrt{a}, & b, & c, & d, & \frac{a}{d}, & \{x_k,y_k\} \\ \sqrt{a}, & -\sqrt{a}, & \frac{qa}{b}, & \frac{qa}{c}, & \frac{qa}{d}, & qd, & \{\frac{qa}{x_k},\frac{qa}{y_k}\} \end{matrix}; \frac{q^{1-N}a}{bc}\right] \\
&(3.1b)\qquad = (a/d)^N\left[\begin{matrix} q,q,qa,\frac{q}{a},\frac{qa}{bd},\frac{qa}{cd},\frac{qd}{b},\frac{qd}{c} \\ \frac{qa}{d},\frac{qd}{a},qd,\frac{q}{d},\frac{q}{b},\frac{q}{c},\frac{qa}{b},\frac{qa}{c} \end{matrix};q\right]_\infty \prod_{k=1}^p\left[\begin{matrix} \frac{qd}{x_k},\frac{qd}{y_k} \\ \frac{qa}{x_k},\frac{qa}{y_k} \end{matrix};q\right]_{n_k}
\end{aligned}
$$

*provided that $|q^{1-N}a/bc|<1$ and the bilateral series is well-defined.*

**Proof.** Similar to the proof of Theorem 1, consider another meromorphic function in complex variable $z$, with multiplicative reflection property $F(z)=F(a/z)$, defined by

$$
\begin{aligned}
&(3.2a)\qquad F(z) = \left[\begin{matrix} q,\ q,\ a,\ q/a,\ qa/bz,\ qa/cz,\ qz/b,\ qz/c \\ a/z,\ qz/a,\ z,\ q/z,\ q/b,\ q/c,\ qa/b,\ qa/c \end{matrix};q\right]_\infty \times \\
&(3.2b)\qquad\qquad \times (a/z)^N\prod_{k=1}^p\left[\begin{matrix} qz/x_k,\ qz/y_k \\ qa/x_k,\ qa/y_k \end{matrix};q\right]_{n_k}.
\end{aligned}
$$


All rights reserved © Bolyai Institute, University of Szeged

It has simple poles $z=q^j$ and $z=aq^j$ $(j=0,\pm1,\pm2,\cdots)$ with residues

$$\text{(3.3a)}\quad z=q^{-j},-q^{-j(1+N)}\left\{\frac{qa}{bc}\right\}^j\left[\begin{matrix}b, & c\\ qa/b, & qa/c\end{matrix};q\right]_j\prod_{k=1}^{p}\left[\begin{matrix}x_k, & y_k\\ qa/x_k, & qa/y_k\end{matrix};q\right]_j$$

$$\text{(3.3b)}\quad z=aq^j,aq^{j(1-N)}\left\{\frac{qa}{bc}\right\}^j\left[\begin{matrix}b, & c\\ qa/b, & qa/c\end{matrix};q\right]_j\prod_{k=1}^{p}\left[\begin{matrix}x_k, & y_k\\ qa/x_k, & qa/y_k\end{matrix};q\right]_j.$$

Write $|a|=|q|^A$ and $A=\varepsilon \pmod 1$. Denote by $\delta_m$ and $\partial_m$ the circles with the center at the origin and radii $|q|^{m+(1+\varepsilon)/2}$ and $|q|^{-m+(1+\varepsilon)/2}$, respectively. It is clear that there is no pole of $F(z)$ passing through $\delta_m$ or $\partial_m$. Then the residue theorem (cf. [9, §3.1]) states that

$$\text{(3.4)}\quad \frac{F(z)}{z}+\sum_{\tau}\frac{\operatorname{Res}\left[F(z)\right]_{z=\tau}}{\tau\,(\tau-z)}=\frac{1}{2\pi i}\int_{\partial_m}\frac{F(t)}{t\,(t-z)}\mathrm{dt}-\frac{1}{2\pi i}\int_{\delta_m}\frac{F(t)}{t\,(t-z)}\mathrm{dt}$$

where the summation runs over all poles $\{\tau\}$ of $F(z)$ between contours $\delta_m$ and $\partial_m$. For sufficiently large $m$, we can estimate that

$$\text{(3.5a)}\quad \int_{\delta_m}\frac{F(t)}{t\,(t-z)}\mathrm{dt}=O\left\{q^{-mN}\left[\begin{matrix}q^{-m+(1-\varepsilon)/2}a/b, & q^{-m+(1-\varepsilon)/2}a/c\\ q^{-m+(1-\varepsilon)/2}, & q^{-m-(1+\varepsilon)/2}a\end{matrix};q\right]_m\right\}$$

$$\text{(3.5b)}\quad =O\left\{(q^{1-N}a/bc)^m\right\}$$

$$\text{(3.5c)}\quad \int_{\partial_m}\frac{F(t)}{t\,(t-z)}\mathrm{dt}=O\left\{q^{m-mN}\left[\begin{matrix}q^{-m+(3+\varepsilon)/2}/b, & q^{-m+(3+\varepsilon)/2}/c\\ q^{-m+(1+\varepsilon)/2}, & q^{-m+(3+\varepsilon)/2}/a\end{matrix};q\right]_m\right\}$$

$$\text{(3.5d)}\quad =O\left\{(q^{2-N}a/bc)^m\right\}.$$

When $|q^{1-N}a/bc|<1$, both integrals of (3.4) tend to zero, which coincides with the convergent condition of bilateral series (3.1a). Letting $m\to\infty$, we may rewrite (3.4), with $z$ being replaced by $d$, as the basic very well-poised bilateral hypergeometric formula stated in the Theorem 2. Unfortunately, this very well-poised evaluation is not a proper extension of (1.6), even though we have expected that.

When $N=0$, it reduces to a special case of the Bailey formula (cf. [8, §7.1])

$$\text{(3.6a)}\quad {}_6\psi_6\left[\begin{matrix}q\sqrt{a}, & -q\sqrt{a}, & b, & c, & d, & e\\ \sqrt{a}, & -q\sqrt{a}, & qa/b, & qa/c, & qa/d, & qa/e\end{matrix};qa^2/bcde\right]$$

$$\text{(3.6b)}\quad =\left[\begin{matrix}q,\,qa,\,q/a,\,qa/bc,qa/bd,\,qa/be,\,qa/cd,\,qa/ce,\,qa/de\\ qa^2/bcde,\,qa/b,\,qa/c,\,qa/d,\,qa/e,\,q/b,\,q/c,\,q/d,\,q/e\end{matrix};q\right]_\infty$$

where $|qa^2/bcde|<1$.


All rights reserved © Bolyai Institute, University of Szeged

## 4. The terminating exceptional cases

For four complex sequences $\{a_k\}$, $\{b_k\}$, $\{c_k\}$ and $\{d_k\}$ with $b_k\,d_k \neq 0$, define two polynomials by

$$P(z) = \prod_{k=1}^{m} (a_k + zb_k) \tag{4.1}$$

$$Q(z) = \prod_{k=1}^{n} (c_k + zd_k). \tag{4.2}$$

If $(c_1/d_1,\, c_2/d_2, \cdots,\, c_n/d_n)$ are distinct, we can make the expansion in partial fractions

$$\frac{P(z)}{Q(z)} = R(z) + \sum_{k=1}^{n} \frac{w_k}{c_k + zd_k} \tag{4.3a}$$

$$w_k = \frac{\prod_{i=1}^{m}(a_i - b_i c_k/d_k)}{\prod_{j=1,\ j\neq k}^{n}(c_j - d_j c_k/d_k)} \tag{4.3b}$$

where $R(z)$ is a polynomial of degree $m-n$ in $z$.

Denote $(b_1 b_2 \cdots b_m)/(d_1 d_2 \cdots d_n)$ by $B/D$ and the elementary symmetric functions of the first degree in $\{a_k/b_k\}$, $\{c_k/d_k\}$, $\{{a_k}^2/{b_k}^2\}$ and $\{{c_k}^2/{d_k}^2\}$, respectively, by $\sigma(a/b)$, $\sigma(c/d)$, $\sigma(a^2/b^2)$ and $\sigma(c^2/d^2)$. For $m < n$, we have directly

$$R(z) = \lim_{z\to\infty} P(z)/Q(z) = 0.$$

While for $m \geq n$, write

$$R(z) = \alpha z^{m-n} + \beta z^{m-n-1} + \gamma z^{m-n-2} + \cdots. \tag{4.4}$$

We may compute the first few terms of $R(z)$, in succession, by

$$\alpha = \lim_{z\to\infty} z^{n-m}\frac{P(z)}{Q(z)} = B/D, \quad (m-n \geq 0)$$

$$\beta = \lim_{z\to\infty} z^{1+n-m}\left\{\frac{P(z)}{Q(z)} - \alpha z^{m-n}\right\}, \quad (m-n \geq 1)$$

$$\gamma = \lim_{z\to\infty} z^{2+n-m}\left\{\frac{P(z)}{Q(z)} - \alpha z^{m-n} - \beta z^{m-n-1}\right\}, \quad (m-n \geq 2).$$

Their explicit expressions are summarized in the following


All rights reserved © Bolyai Institute, University of Szeged

**Lemma 3.**

$$R(z) = 0, \quad (m < n) \tag{4.5a}$$

$$R(z) = \alpha, \quad (m = n) \tag{4.5b}$$

$$R(z) = \alpha z + \beta, \quad (m = n+1) \tag{4.5c}$$

$$R(z) = \alpha z^2 + \beta z + \gamma, \quad (m = n+2) \tag{4.5d}$$

*where*

$$\alpha = B/D \tag{4.6a}$$

$$\beta = B/D\,\{\sigma(a/b) - \sigma(c/d)\} \tag{4.6b}$$

$$\gamma = B/D\left\{\frac{[\sigma(a/b) - \sigma(c/d)]^2}{2} - \frac{\sigma(a^2/b^2) - \sigma(c^2/d^2)}{2}\right\}. \tag{4.6c}$$

For complex numbers $\{x_k, y_k\}_{k=1}^p$ and nonnegative integers $\{n_k\}_{k=1}^p$ with $x_k + y_k = 1 + a + n_k$ $(k = 1, 2, \ldots, p)$ and $N = \sum_{k=1}^p n_k$, define two polynomials by

$$P(z) = \prod_{k=1}^{p} (1 + z - x_k)_{n_k}\,(1 + z - y_k)_{n_k} \tag{4.7a}$$

$$Q(z) = (z)_{m+1}\,(a - z)_{m+1}. \tag{4.7b}$$

By means of (4.3), it is easy to derive

$${}_{2p+5}F_{4+2p}\left[\begin{matrix} a, & 1 + a/2, & z, & a - z, & -m, & \{x_k, y_k\} \\ & a/2, & z^*, & 1 + z, & 1 + a + m, & \{x_k^*, y_k^*\} \end{matrix}\right] \tag{4.8a}$$

$$= \left\{\frac{P(z)}{Q(z)} - R(z)\right\} \frac{m!z\,(a - z)(1 + a)_m}{\prod_{k=1}^p (1 + a - x_k)_{n_k}\,(1 + a - y_k)_{n_k}} \tag{4.8b}$$

where $R(z)$ is a polynomial of degree $2(N - m - 1)$ in $z$.

The symmetric functions may be computed as follows:

$$B/D = (-1)^{m+1} \tag{4.9a}$$

$$\sigma(a/b) = \sum_{k=1}^{p} \sum_{i=1}^{n_k} (2i - x_k - y_k) \tag{4.9b}$$

$$= \sum_{k=1}^{p} n_k(1 + n_k - x_k - y_k) = -aN \tag{4.9c}$$


All rights reserved © Bolyai Institute, University of Szeged

$$\sigma(c/d)=\sum_{j=0}^{m}(k-a-k)=-a(m+1) \tag{4.9d}$$

$$\sigma(a^2/b^2)=\sum_{k=1}^{p}\sum_{i=1}^{n_k}\{(i-x_k)^2+(i-y_k)^2\} \tag{4.9e}$$

$$=\sum_{k=1}^{p}\left\{\binom{2+2n_k}{3}/2+n_k(x_k^2+y_k^2)-n_k(1+n_k)(x_k+y_k)\right\} \tag{4.9f}$$

$$=\sum_{k=1}^{p}\left\{\binom{2+2n_k}{3}/2+n_k(a^2+a+an_k-2x_ky_k)\right\} \tag{4.9g}$$

$$=a^2N+\sum_{k=1}^{p}\left\{\frac{2}{3}(1+2n_k+3a)\binom{1+n_k}{2}-2n_kx_ky_k\right\} \tag{4.9h}$$

$$\sigma(c^2/d^2)=\sum_{j=0}^{m}\{j^2+(a+j)^2\} \tag{4.9i}$$

$$=2\binom{m+1}{3}+\binom{m+1}{2}+2\binom{a+m+1}{3}+\binom{a+m+1}{2} \tag{4.9j}$$

$$=a(a+m)(1+m)+\binom{2m+2}{3}/2. \tag{4.9k}$$

In view of the Lemma 3, we have the following very well-poised terminating formulas.

**Theorem 4.** *For complex numbers $a$, $c$ and $\{x_k,\ y_k\}_{k=1}^{p}$ satisfying $x_k+y_k=1+a+n_k$ $(k=1,2,\ldots,p)$ and $N=\sum_{k=1}^{p}n_k$ with $m$ and $\{n_k\}_{k=1}^{p}$ being nonnegative integers, there holds*

$$ {}_{2p+5}F_{4+2p}\left[\begin{matrix}a, & 1+\frac{a}{2}, & c, & a-c, & -m, & \{x_k,y_k\}\\ & \frac{a}{2}, & c^*, & 1+c, & 1+a+m, & \{x_k^*,y_k^*\}\end{matrix}\right] \tag{4.10a}$$

$$=\left[\begin{matrix}1, 1+a\\ 1+c, 1+a-c\end{matrix}\right]_m\prod_{k=1}^{p}\left[\begin{matrix}1+c-x_k, & 1+c-y_k\\ 1+a-x_k, & 1+a-y_k\end{matrix}\right]_{n_k} \tag{4.10b}$$

$$+(-1)^mR(c)\frac{m!c(a-c)(1+a)_m}{\prod_{k=1}^{p}(1+a-x_k)_{n_k}(1+a-y_k)_{n_k}} \tag{4.10c}$$

*where $R(c)$ is a polynomial of degree $2(N-m-1)$ in $c$ whose first few terms are displayed as*

$$m\geq N, R(c)=0 \tag{4.11a}$$


All rights reserved © Bolyai Institute, University of Szeged

$$\text{(4.11b)}\quad m=N-1, R(c)=1$$

$$\text{(4.11c)}\quad m=N-2, R(c)=c^2-ac+\frac{1+2m+3a}{3}\binom{1+m}{2}$$

$$\text{(4.11d)}\qquad +\sum_{k=1}^{p}\left\{n_kx_ky_k-\frac{1+2n_k+3a}{3}\binom{1+n_k}{2}\right\}.$$

In fact, (4.11a) corresponds to the terminating case of the bilateral formula stated in Theorem 1. When $p=1$, the evaluations corrersponding to (4.11) may be recovered from the Whipple transformation (cf. [1, p.25])

$$\text{(4.12a)}\quad {}_7F_6\left[\begin{matrix}a, & 1+a/2, & b, & c, & d, & e, & -n\\ & a/2, & b^*, & c^*, & d^*, & e^*, & 1+a+n\end{matrix}\right]$$

$$\text{(4.12b)}\quad =\left[\begin{matrix}1+a, & 1+a-b-c\\ 1+a-b, & 1+a-c\end{matrix}\right]_n {}_4F_3\left[\begin{matrix}-n, b, c, & 1+a-d-e\\ d^*, e^*, & b+c-a-n\end{matrix}\right]$$

and the Dougall–Dixon formula (cf. [1, p.26])

$$\text{(4.13a)}\quad {}_7F_6\left[\begin{matrix}a, & 1+a/2, & b, & c, & d, & e, & -n\\ & a/2, & b^*, & c^*, & d^*, & e^*, & 1+a+n\end{matrix}\right]$$

$$\text{(4.13b)}\quad =\left[\begin{matrix}1+a, & 1+a-b-c, & 1+a-b-d, & 1+a-c-d\\ 1+a-b, & 1+a-c, & 1+a-d, & 1+a-b-c-d\end{matrix}\right]_n$$

provided that $1+2a+n=b+c+d+e$ for the latter.

Now we define another rational function by

$$G(z)=\frac{(a/z)^N}{(z;q)_{m+1}(a/z;q)_{m+1}}\prod_{k=1}^{p}(qz/x_k;\ q)_{n_k}(qz/y_k;\ q)_{n_k}.$$

The corresponding partial fraction expansion yields

$$\text{(4.14a)}\quad {}_{2p+6}\phi_{5+2p}\left[\begin{matrix}a, & q\sqrt{a}, -q\sqrt{a}, & z, \frac{a}{z}, & q^{-m}, & \{x_k, y_k\}\\ & \sqrt{a}, -\sqrt{a}, & \frac{qa}{z}, qz, & q^{1+m}a, & \{\frac{qa}{x_k}, \frac{qa}{y_k}\}\end{matrix}; q^{1+m-N}\right]$$

$$\text{(4.14b)}\quad =\{G(z)-R(z)\}\frac{(1-z)\,(1-a/z)\,(q;q)_m\,(qa;q)_m}{\prod_{k=1}^{p}[qa/x_k,\ qa/y_k;\ q]_{n_k}}$$

where $R(z)$ is a finite Laurent polynomial of degree $(N-m-1)$ with reflection property $R(z)=R(a/z)$ on account of $G(z)=G(a/z)$. Then we may rewrite (4.4) as

$$\text{(4.15)}\qquad R(z)=\alpha[z^{N-m-1}+(a/z)^{N-m-1}]+$$

$$\text{(4.16)}\qquad +\beta[z^{N-m-2}+(a/z)^{N-m-2}]+$$

$$\text{(4.17)}\qquad +\gamma[z^{N-m-3}+(a/z)^{N-m-3}]+\cdots.$$


All rights reserved © Bolyai Institute, University of Szeged

Noting that

$$G(z)=(-1)^{m+1}q^{-\binom{m+1}{2}}(a/z)^{N-m-1}\frac{\prod_{k=1}^{p}[qz/x_k,\ qz/y_k;\ q]_{n_k}}{(z;q)_{m+1}(q^{-m}z/a;q)_{m+1}} \tag{4.18}$$

we may display the computation of symmetric sums.

$$B/D=(-1)^{m+1}q^{-\binom{m+1}{2}}a^{N-m-1}\prod_{k=1}^{p}\{q^{1+n_k}/x_ky_k\}^{n_k}/a^{-m-1} \tag{4.19a}$$

$$=(-1)^{m+1}q^{-\binom{m+1}{2}} \tag{4.19b}$$

$$\sigma(a/b)=-\sum_{k=1}^{p}\sum_{i=1}^{n_k}q^{-i}\{x_k+y_k\} \tag{4.19c}$$

$$=\sum_{k=1}^{p}\frac{1-qa/x_ky_k}{1-q}(x_k+y_k) \tag{4.19d}$$

$$\sigma(c/d)=-\sum_{j=0}^{m}q^{-j}(1+q^ma) \tag{4.19e}$$

$$=-q^{-m}\frac{1-q^{1+m}}{1-q}(1+q^ma) \tag{4.19f}$$

$$\sigma(a^2/b^2)=\sum_{k=1}^{p}\sum_{i=1}^{n_k}q^{-2i}\{x_k^2+y_k^2\} \tag{4.19g}$$

$$=-\sum_{k=1}^{p}\frac{1-q^2a^2/x_k^2y_k^2}{1-q^2}(x_k^2+y_k^2) \tag{4.19h}$$

$$\sigma(c^2/d^2)=\sum_{j=0}^{m}q^{-2j}(1+q^{2m}a^2) \tag{4.19i}$$

$$=q^{-2m}\frac{1-q^{2+2m}}{1-q^2}(1+q^{2m}a^2). \tag{4.19j}$$

Combining these results with Lemma 3, we get the $q$-analogue of Theorem 4 as follows.

**Theorem 5.** *For complex numbers $a$, $c$ and $\{x_k,y_k\}_{k=1}^{p}$ satisfying $x_k\cdot y_k=aq^{1+n_k}$ $(k=1,2,\cdots,p)$ and $N=\sum_{k=1}^{p}n_k$ with $m$ and $\{n_k\}_{k=1}^{p}$ being nonnegative integers, there holds*

$$ {}_{2p+6}\phi_{5+2p}\left[\begin{matrix}a, & q\sqrt{a},-q\sqrt{a}, & c,\frac{a}{c}, & q^{-m}, & \{x_k,y_k\}\\ & \sqrt{a},-\sqrt{a}, & \frac{qa}{c},qc, & q^{1+m}a, & \{\frac{qa}{x_k},\frac{qa}{y_k}\}\end{matrix};q^{1+m-N}\right] \tag{4.20a}$$


All rights reserved © Bolyai Institute, University of Szeged

$$= (a/c)^N \left[ \begin{matrix} q, qa \\ qc, qa/c \end{matrix} ; q \right]_m \prod_{k=1}^{p} \left[ \begin{matrix} qc/x_k,\ qc/y_k \\ qa/x_k,\ qa/y_k \end{matrix} ; q \right]_{n_k} + \tag{4.20b}$$

$$+ R(c)\ (-1)^m\ q^{-\binom{m+1}{2}} \frac{(1-c)\,(1-a/c)\,(q;q)_m (qa;q)_m}{\prod_{k=1}^{p} [qa/x_k,\ qa/y_k;\ q]_{n_k}} \tag{4.20c}$$

*where $R(c)$ is a Laurent polynomial of degree $(N-m-1)$ in $c$ whose first few terms are displayed, in terms of the symmetric functions from* (4.19)*, as*

$$m \geq N, R(c) = 0 \tag{4.21a}$$

$$m = N-1,\ R(c) = 1 \tag{4.21b}$$

$$m = N-2,\ R(c) = (c + a/c) + q^{-m} \frac{1-q^{1+m}}{1-q} (1 + q^m a) + \tag{4.21c}$$

$$+ \sum_{k=1}^{p} \frac{1 - qa/x_k y_k}{1-q} (x_k + y_k) \tag{4.21d}$$

$$m = N-3,\ R(c) = (c^2 + a^2/c^2) + (c + a/c)[\sigma(a/b) - \sigma(c/d)] + \tag{4.21e}$$

$$+ \frac{1}{2} \{ [\sigma(a/b) - \sigma(c/d)]^2 - [\sigma(a^2/b^2) - \sigma(c^2/d^2)] \}. \tag{4.21f}$$

Similarly, the terminating case of the bilateral formula stated in Theorem 2 reduces to (4.21a). When $p = 1$, the evaluations corresponding to (4.21) are particular cases of Watson's $q$-analogue of the Whipple transformation (cf. [8, §8.5])

$${}_8\phi_7 \left[ \begin{matrix} a, & q\sqrt{a}, & -q\sqrt{a}, & b, & c, & d, & e, & q^{-n} \\ & \sqrt{a}, & -\sqrt{a}, & qa/b, & qa/c, & qa/d, & qa/e, & q^{1+n}a \end{matrix} ; \frac{q^{2+n}a^2}{bcde} \right] \tag{4.22a}$$

$$= \left[ \begin{matrix} qa, & qa/bc \\ qa/b, & qa/c \end{matrix} ; q \right]_n {}_4\phi_3 \left[ \begin{matrix} q^{-n}, & b, & c, & qa/de \\ & qa/d, & qa/e, & q^{-n}bc/a \end{matrix} ; q \right] \tag{4.22b}$$

and Jackson's $q$-analogue of the Dougall–Dixon theorem (cf. [8, §8.3])

$${}_8\phi_7 \left[ \begin{matrix} a, & q\sqrt{a}, & -q\sqrt{a}, & b, & c, & d, & e, & q^{-n} \\ & \sqrt{a}, & -\sqrt{a}, & qa/b, & qa/c, & qa/d, & qa/e, & q^{1+n}a \end{matrix} ; q \right] \tag{4.23a}$$

$$= \left[ \begin{matrix} qa,\ qa/bc,\ qa/bd,\ qa/cd \\ qa/b,\ qa/c,\ qa/d,\ qa/bcd \end{matrix} ; q \right]_n \tag{4.23b}$$

where $bcde = q^{1+n}a^2$ for the latter.


All rights reserved © Bolyai Institute, University of Szeged

## 5. Inverse series relations and the dual formulae

For an inverse pair of infinite upper triangular matrices

$$A = (a_{ij})_{0\le i,\, j<\infty} \tag{5.1a}$$

$$B = (b_{ij})_{0\le i,\, j<\infty} \tag{5.1b}$$

the system of equations

$$f(n) = \sum_{j=0}^{m} a_{jn} g(j), \qquad n = 0, 1, 2, \ldots \tag{5.2a}$$

is equivalent to the system

$$g(n) = \sum_{j=0}^{m} b_{jn} f(j), \qquad n = 0, 1, 2, \ldots. \tag{5.2b}$$

If one member of the inverse pair can be properly specified as a known relation, the other will provide a dual relation. That is the so called inversion technique, which has been used efficiently to systematize combinatorial identities [3]. Based on two very well-poised terminating identities established in the last section, we will pursue their dual formulae through the inversion technique.

For $n \equiv r \pmod p$, define $p$-nonnegative integers by

$$n_k := \begin{cases} [n/p], & k + r < p \\ 1 + [n/p], & k + r \ge p \end{cases} \tag{5.3a}$$

$$n + 1 = \sum_{k=1}^{p} n_k. \tag{5.3b}$$

It is easy to verify that $\omega$ defined by

$$\left.\begin{array}{ll} \omega(k) = p - r - k, & (r + k < p) \\ \omega(k) = 2p - r - k, & (r + k \ge p) \end{array}\right\} \quad (k = 1, 2, \ldots, p) \tag{5.4}$$

is a permutation on $\{1, 2, \cdots, p\}$. Let

$$\left.\begin{array}{l} x_k = b + \frac{n+k}{p} \\ y_k = a - b + \frac{\omega(k)}{p} \end{array}\right\} \quad (k = 1, 2, \ldots, p). \tag{5.5}$$

Then we have

$$x_k + y_k = 1 + a + n_k, \quad (k = 1, 2, \ldots, p) \tag{5.6a}$$


All rights reserved © Bolyai Institute, University of Szeged

$$\{x_k\}_{k=1}^{p} = \Big\{b + \frac{n+k}{p}\Big\}_{k=1}^{p} \tag{5.6b}$$

$$\{y_k\}_{k=1}^{p} = \Big\{a - b + \frac{k}{p}\Big\}_{k=1}^{p} \tag{5.6c}$$

$$\{1+a-x_k\}_{k=1}^{p} = \Big\{1 + a - b - \frac{n+k}{p}\Big\}_{k=1}^{p} \tag{5.6d}$$

$$\{1+a-y_k\}_{k=1}^{p} = \Big\{1 + b - \frac{k}{p}\Big\}_{k=1}^{p} \tag{5.6e}$$

and the corresponding product of shifted factorials

$$\prod_{k=1}^{p} \begin{bmatrix} 1+c-x_k, 1+c-y_k \\ 1+a-x_k, 1+a-y_k \end{bmatrix}_{n_k} \tag{5.7a}$$

$$= \begin{bmatrix} bp-cp, & bp+cp-ap \\ bp, & bp-ap \end{bmatrix}_{n+1}. \tag{5.7b}$$

For $n = m-1$, the formula corresponding to (4.11a) in Theorem 4 may be reformulated, under the above displayed parameter setting, as

$${}_{2p+5}F_{4+2p}\left[\begin{matrix} a, & 1+a/2, & c, & a-c, & -m, & \{b+\frac{k-1+m}{p}, a-b+\frac{k}{p}\} \\ & a/2, & c^*, & 1+c, & 1+a+m, & \{b^*-\frac{k-1+m}{p}, 1+b-\frac{k}{p}\} \end{matrix}\right] \tag{5.8a}$$

$$= \begin{bmatrix} 1, 1+a \\ 1+c, 1+a-c \end{bmatrix}_m \begin{bmatrix} bp-cp, bp+cp-ap \\ bp, bp-ap \end{bmatrix}_m. \tag{5.8b}$$

By means of

$$\prod_{k=1}^{p} \begin{bmatrix} b+\frac{k-1+m}{p}, a-b+\frac{k}{p} \\ 1+a-b-\frac{k-1+m}{p}, 1+b-\frac{k}{p} \end{bmatrix}_j \tag{5.9a}$$

$$= \begin{bmatrix} bp+m, 1+ap-bp \\ 1+ap-bp-m, bp \end{bmatrix}_{pj} \tag{5.9b}$$

$$= \begin{bmatrix} bp+pj, bp-ap-pj \\ bp, bp-ap \end{bmatrix}_m \tag{5.9c}$$

we may restate (5.8) in the following sum

$$\sum_{j=0}^{m} (-1)^j \binom{m}{j} (bp+pj)_m (bp-ap-pj)_m \frac{a+2j}{(a+m)_{j+1}} \frac{c}{c+j} \frac{a-c}{a-c+j} (a)_j \tag{5.10a}$$

$$= \begin{bmatrix} 1, a \\ 1+c, 1+a-c \end{bmatrix}_m (bp-cp)_m (bp+cp-ap)_m. \tag{5.10b}$$


All rights reserved © Bolyai Institute, University of Szeged

Recall a pair of inverse series relations [3, §4]

$$f(m)=\sum_{j=0}^{m}(-1)^j\binom{m}{j}\varphi(t+j;m)\varphi(-j;m)\frac{t+2j}{(t+m)_{j+1}}g(j) \tag{5.11a}$$

$$g(m)=\sum_{j=0}^{m}(-1)^j\binom{m}{j}\frac{a_j+(t+j)b_j}{\varphi(t+m;j+1)}\frac{a_j-jb_j}{\varphi(-m;j+1)}(t+j)_m f(j) \tag{5.11b}$$

where $\varphi(z;n)$ is defined by

$$\varphi(z;0)\equiv 1,\qquad \varphi(z;n)=\prod_{k=0}^{n-1}(a_k+zb_k), n=1,2,\dots \tag{5.12}$$

as a polynomial of degree $n$ in $z$ for two complex sequences $\{a_k,b_k\}_{k\geq 0}$.

If we specify

$$\varphi(z;k)=(bp-ap+pz)_k : t=a \tag{5.13a}$$

$$f(k)=\left[\begin{matrix}1,a\\1+c,1+a-c\end{matrix}\right]_k(bp-cp)_k(bp+cp-ap)_k \tag{5.13b}$$

$$g(k)=\frac{c}{c+k}\frac{a-c}{a-c+k}(a)_k. \tag{5.13c}$$

Then (5.10) matches to (5.11a) perfectly. The dual formulae from (5.11b) reads as

$$\sum_{j=0}^{m}(-1)^j\binom{m}{j}\frac{bp+pj+j}{(bp+pm)_{j+1}}\frac{bp-ap-pj+j}{(bp-ap-pm)_{j+1}}(a+j)_m\times \tag{5.14a}$$

$$\times\left[\begin{matrix}1,a\\1+c,1+a-c\end{matrix}\right]_j(bp-cp)_j(bp+cp-ap)_j \tag{5.14b}$$

$$=\frac{c\,(a-c)}{(c+m)\,(a-c+m)}(a)_m \tag{5.14c}$$

which may be stated in terms of hypergeometric evaluation (cf. [2, Eq. 5.3])

$${}_7F_6\left[\begin{matrix}1,\ 1+\frac{bp}{p+1},\ 1+\frac{ap-bp}{p-1},\ p(b-c),\ p(b-a+c),\ a+m,\ -m\\ \frac{bp}{p+1},\ \frac{ap-bp}{p-1},\ 1+c,\ c^*,\ 1+p(b-a-m),\ 1+p(b+m)\end{matrix}\right] \tag{5.15a}$$

$$=\frac{c}{c+m}\frac{a-c}{a-c+m}\frac{b+m}{b}\frac{a-b+m}{a-b}. \tag{5.15b}$$

Assume the nonnegative integers $\{n_1,n_2,\dots,n_p\}$ and the permutation $\omega$ on $\{1,2,\dots,p\}$ defined by (5.3) and (5.4) respectively. Let

$$x_k=q^{\frac{n+k}{p}}b \tag{5.16a}$$

$$y_k=q^{\frac{\omega(k)}{p}}a/b. \tag{5.16b}$$


All rights reserved © Bolyai Institute, University of Szeged

Then we have

$$x_k y_k = q^{1+n_k} a, (k = 1, 2, \cdots, p) \tag{5.17a}$$

$$\{x_k\}_{k=1}^p = \{q^{\frac{n+k}{p}} b\}_{k=1}^p \tag{5.17b}$$

$$\{y_k\}_{k=1}^p = \{q^{\frac{k}{p}} a/b\}_{k=1}^p \tag{5.17c}$$

$$\{qa/x_k\}_{k=1}^p = \{q^{\frac{p-n-k}{p}} a/b\}_{k=1}^p \tag{5.17d}$$

$$\{qa/y_k\}_{k=1}^p = \{q^{\frac{p-k}{p}} /b\}_{k=1}^p \tag{5.17e}$$

and the corresponding product of shifted factorials

$$\prod_{k=1}^{p} \left[ \begin{matrix} qc/x_k, & qc/y_k \\ qa/x_k, & qa/y_k \end{matrix} ; q \right]_{n_k} \tag{5.18a}$$

$$= \left[ \begin{matrix} b/c, & bc/a \\ b, & b/a \end{matrix} ; q^{1/p} \right]_{n+1} (c/a)^{n+1}. \tag{5.18b}$$

For $n = m - 1$, the above displayed parameter setting may be used to rewrite the formula corresponding to (4.21a) in Theorem 5, as

$${}_{2p+6}\phi_{5+2p} \left[ \begin{matrix} a, & q\sqrt{a}, & -q\sqrt{a}, & c, & \frac{a}{c}, & q^{-m}, & \{q^{\frac{m+k-1}{p}} b q^{\frac{k}{p}} \frac{a}{b}\} \\ & \sqrt{a}, & -\sqrt{a}, & \frac{qa}{c}, & qc, & q^{1+m}a, & \{q^{\frac{p-m-k+1}{p}} \frac{a}{b}, q^{\frac{p-k}{p}} b\} \end{matrix} ; q \right] \tag{5.19a}$$

$$= \left[ \begin{matrix} q, qa \\ qc, qa/c \end{matrix} ; q \right]_m \left[ \begin{matrix} b/c, bc/a \\ b, b/a \end{matrix} ; q^{1/p} \right]_m. \tag{5.19b}$$

By means of

$$\prod_{k=1}^{p} \left[ \begin{matrix} q^{\frac{m+k-1}{p}} b, q^{\frac{k}{p}} a/b \\ q^{\frac{p-m-k+1}{p}} a/b, q^{\frac{p-k}{p}} b \end{matrix} ; q \right]_j \tag{5.20a}$$

$$= \left[ \begin{matrix} q^{\frac{m}{p}} b, q^{1/p} a/b \\ b, q^{\frac{1-m}{p}} a/b \end{matrix} ; q^{1/p} \right]_{pj} \tag{5.20b}$$

$$= \left[ \begin{matrix} q^j b, & q^{-j} b/a \\ b, & b/a \end{matrix} ; q^{1/p} \right]_m q^{mj} \tag{5.20c}$$

we may restate (5.19) in the following sum

$$\sum_{j=0}^{m} (-1)^j \left[ \begin{matrix} m \\ j \end{matrix} \right] (q^j b;\, q^{1/p})_m (q^{-j} b/a;\, q^{1/p})_m \frac{1 - aq^{2j}}{(q^m a;\, q)_{j+1}} \times \tag{5.21a}$$

$$\times\, q^{\binom{j+1}{2}} \frac{1-c}{1-cq^j} \frac{1-a/c}{1-q^j a/c} (a;\, q)_j \tag{5.21b}$$

$$= \left[ \begin{matrix} q, qa \\ qc, qa/c \end{matrix} ; q \right]_m (b/c;\, q^{1/p})_m (bc/a;\, q^{1/p})_m. \tag{5.21c}$$


All rights reserved © Bolyai Institute, University of Szeged

The $q$-analogue of (5.11) reads from [3, §5], as

$$f(m)=\sum_{j=0}^{m}(-1)^j\begin{bmatrix}m\\ j\end{bmatrix}\varphi(tq^j;m)\varphi(q^{-j};m)\frac{1-tq^{2j}}{(q^mt;\,q)_{j+1}}g(j) \tag{5.22a}$$

$$g(m)=\sum_{j=0}^{m}(-1)^j\begin{bmatrix}m\\ j\end{bmatrix}q^{\binom{m-j}{2}}\frac{a_j+tq^jb_j}{\varphi(q^mt;j+1)}\frac{a_j+q^{-j}b_j}{\varphi(q^{-m};j+1)}(q^jt;\,q)_mf(j) \tag{5.22b}$$

where $\varphi(z;n)$ is defined by (5.12) as before.

Notice that (5.22a) specifies, by

$$\varphi(z;k)=(zb/a;\,q^{1/p})_k:t=a \tag{5.23a}$$

$$f(k)=\left[\begin{matrix}q,qa\\ qc,qa/c\end{matrix};q\right]_k(b/c;\,q^{1/p})_k(bc/a;\,q^{1/p})_k \tag{5.23b}$$

$$g(k)=q^{\binom{k+1}{2}}\frac{1-c}{1-cq^k}\frac{1-a/c}{1-q^ka/c}(a;\,q)_k \tag{5.23c}$$

exactly to (5.21), we get the dual formulae accordingly from (5.22b)

$$\sum_{j=0}^{m}(-1)^j\begin{bmatrix}m\\ j\end{bmatrix}q^{\binom{m-j}{2}}\frac{1-bq^{j+j/p}}{(q^mb;\,q^{1/p})_{j+1}}\frac{1-q^{-j+j/p}b/a}{(q^{-m}b/a;\,q^{1/p})_{j+1}}(q^ja;\,q)_m\times \tag{5.24a}$$

$$\times\left[\begin{matrix}q,\,a\\ qc,qa/c\end{matrix};q\right]_j(b/c;\,q^{1/p})_j(bc/a;\,q^{1/p})_j \tag{5.24b}$$

$$=q^{\binom{m+1}{2}}\frac{(1-c)(1-a/c)}{(1-cq^m)(1-q^ma/c)}(a;\,q)_m \tag{5.24c}$$

which may be stated, with $q^{1/p}$ being replaced by $p$, as a bi-basic summation formula (cf. [2, Eq. 3.3])

$$\sum_{j=0}^{m}q^j\frac{1-bp^jq^j}{1-b}\frac{1-p^jq^{-j}b/a}{1-b/a}\left[\begin{matrix}b/c,bc/a\\ bpq^m,pq^{-m}b/a\end{matrix};p\right]_j\left[\begin{matrix}q^{-m},q^ma\\ qc,qa/c\end{matrix};q\right]_j \tag{5.25a}$$

$$=\frac{1-c}{1-cq^m}\frac{1-a/c}{1-q^ma/c}\frac{1-bq^m}{1-b}\frac{1-q^ma/b}{1-a/b}. \tag{5.25b}$$

## References

[1] W. N. Bailey, *Generalized Hypergeometric Series*, Cambridge University Press, Cambridge, 1935.


All rights reserved © Bolyai Institute, University of Szeged

[2] W. C. Chu, An Algebraic Identity and Its Application: The unification of some matrix inverse pairs, *Pure Math. Appl.*, **4** (1993), 175–190.
[3] W. C. Chu, Inversion techniques and combinatorial identities, *Boll. Un. Math. Ital.*, **7** (1993), 737–760.
[4] W. C. Chu, Partial fractions and bilateral summations, *J. Math. Phys.*, **35** (1994), 2036–2042; Erratum: *ibid* **36** (1995), 5198–5199.
[5] G. Gasper, Summation formulas for basic hypergeometric series, *SIAM J. Math. Anal.*, **12** (1981), 196–200.
[6] Per W. Karlsson, Hypergeometric formulas with integral parameter differences, *J. Math. Phys.*, **12** (1971), 270–271.
[7] B. M. Minton, Generalized hypergeometric function of unit argument, *J. Math. Phys.*, **11** (1970), 1375–1376.
[8] L. J. Slater, *Generalized Hypergeometric Functions*, Cambridge University Press, Cambridge, 1966.
[9] E. T. Titchmarsh, *The Theory of Functions*, Oxford University Press, Oxford, 1939.

W. Chu, 9 rue du Regard, 75006 Paris, France; *e-mail*: chu.dicastro@wanadoo.fr


All rights reserved © Bolyai Institute, University of Szeged